Fady Youssef
10/27/2025
CPSC 321

Homework 5

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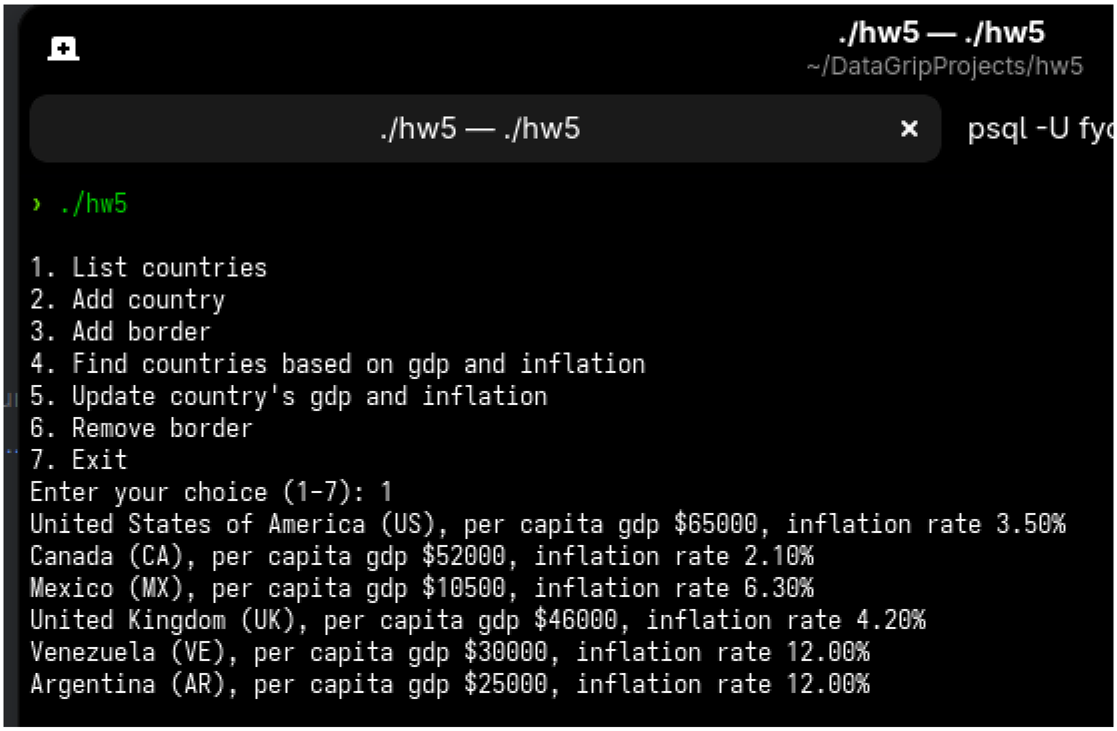
1. 

   Successfully listing countries

```
./hw5 — ./hw5
~/DataGripProjects/hw5
./hw5 — ./hw5    psql -U fyoussef -h cps-postgres

1. List countries
2. Add country
3. Add border
4. Find countries based on gdp and inflation
5. Update country's gdp and inflation
6. Remove border
7. Exit
Enter your choice (1-7): 2
Country code..................: FR
Country name..................: France
Country per capita gdp (USD)..: 45000
Country inflation (pct).......: 2.5
Country France (FR) added.

1. List countries
2. Add country
3. Add border
4. Find countries based on gdp and inflation
5. Update country's gdp and inflation
6. Remove border
7. Exit
Enter your choice (1-7): 1
United States of America (US), per capita gdp $65000, inflation rate 3.50%
Canada (CA), per capita gdp $52000, inflation rate 2.10%
Mexico (MX), per capita gdp $10500, inflation rate 6.30%
United Kingdom (UK), per capita gdp $46000, inflation rate 4.20%
Venezuela (VE), per capita gdp $30000, inflation rate 12.00%
Argentina (AR), per capita gdp $25000, inflation rate 12.00%
France (FR), per capita gdp $45000, inflation rate 2.50%
```

2.

Successfully adding a country to the list of countries.

```
1. List countries
2. Add country
3. Add border
4. Find countries based on gdp and inflation
5. Update country's gdp and inflation
6. Remove border
7. Exit
Enter your choice (1-7): 2
Country code..................: US
Country name..................: USA
Country per capita gdp (USD)..: 1
Country inflation (pct).......: 1
Country with code US already exists.
```

Error showing handling of existing countries.

3.
```
1. List countries
2. Add country
3. Add border
4. Find countries based on gdp and inflation
5. Update country's gdp and inflation
6. Remove border
7. Exit
Enter your choice (1-7): 3
Country code 1..: FR
Country code 2..: UK
Border length...: 50
Border added between FR and UK length 50.
```

Successfully added borders.

```
1. List countries
2. Add country
3. Add border
4. Find countries based on gdp and inflation
5. Update country's gdp and inflation
6. Remove border
7. Exit
Enter your choice (1-7): 3
Country code 1..: ZZ
Country code 2..: US
Border length...: 50
Country ZZ does not exist.
```

Failure adding border for country that doesn’t exist

4.

```
1. List countries
2. Add country
3. Add border
4. Find countries based on gdp and inflation
5. Update country's gdp and inflation
6. Remove border
7. Exit
Enter your choice (1-7): 4
Minimum per capita gdp (USD)..: 20000
Maximum per capita gdp (USD)..: 70000
Minimum inflation (pct).......: 1.0
Maximum inflation (pct).......: 15.0
United States of America (US), per capita gdp $65000, inflation rate 3.50%
Canada (CA), per capita gdp $52000, inflation rate 2.10%
United Kingdom (UK), per capita gdp $46000, inflation rate 4.20%
France (FR), per capita gdp $45000, inflation rate 2.50%
Venezuela (VE), per capita gdp $30000, inflation rate 12.00%
Argentina (AR), per capita gdp $25000, inflation rate 12.00%
```

Successfully showing list of matching countries

5.

```
1. List countries
2. Add country
3. Add border
4. Find countries based on gdp and inflation
5. Update country's gdp and inflation
6. Remove border
7. Exit
Enter your choice (1-7): 5
Country code..................: FR
Country per capita gdp (USD)..: 46000
Country inflation (pct).......: 2.8
Country FR updated to gdp $46000 and inflation 2.80%
```

Successfully updating country

```
1. List countries
2. Add country
3. Add border
4. Find countries based on gdp and inflation
5. Update country's gdp and inflation
6. Remove border
7. Exit
Enter your choice (1-7): 5
Country code..................: GU
Country per capita gdp (USD)..: 1
Country inflation (pct).......: 1
Country GU does not exist.
```

Can’t update a country that doesn’t exist

```
Country GU does not exist.

1. List countries
2. Add country
3. Add border
4. Find countries based on gdp and inflation
5. Update country's gdp and inflation
6. Remove border
7. Exit
Enter your choice (1-7): 6
Country code 1..: FR
Country code 2..: UK
Border between FR and UK removed.
```

6. Successfully removing border

```
Border between FR and UK removed.

1. List countries
2. Add country
3. Add border
4. Find countries based on gdp and inflation
5. Update country's gdp and inflation
6. Remove border
7. Exit
Enter your choice (1-7): 6
Country code 1..: FR
Country code 2..: UK
Border between FR and UK does not exist.
```

The border no longer exists and can't be altered further.

```
1. List countries
2. Add country
3. Add border
4. Find countries based on gdp and inflation
5. Update country's gdp and inflation
6. Remove border
7. Exit
Enter your choice (1-7): 7
Goodbye.
~/DataGripProjects/hw5 ›
```

7. Program successfully exits

Challenges Faced:

1. A major problem I faced that took a significant amount of time was implementing the error handling between the c++ and SQL code. After implementing the helper functions, I had to go further in depth implementing errors and constantly going back and forth to check data types in the SQL code and implementing in the C++ code. This solution also helped me make the program more rigid because I was able to thoroughly go through how all the data would be inputted and adjusted the code accordingly.

2. Another was implementing bi-directional border logic for both adding and removing borders. I added the helper function(borderExists) with a proper WHERE clause that allowed me to check this at once and passed it through both add and remove border functions which fixed the problem. (it helped as I also had this problem in a previous homework).

3. The starter files were a huge help but it took a considerable amount of time understanding how and why certain functions were being used and called then combining them with SQL queries and understanding the ordering of everything.